TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology
Wardha Road, Nagpur - 441108
Accredited with NAAC A+ Grade
Approved by AICTE, New Delhi, Govt. of Maharashtra

(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)
Department of Information Technology
Session 2024-25(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: VI

Programme: B. Tech Information Technology

Course: Artificial Intelligence and Machine Learning

Time: 3 hours **Max Marks:** 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33602	Course Outcome
CO1	Identify knowledge representation, problem solving and learning methods of AI.
CO2	Discuss knowledge and reasoning and its representation in AI.
CO3	Describe artificial intelligence, machine learning and deep learning
CO4	Analyze supervised learning algorithms and decision trees.
CO5	Evaluate machine learning models using cross-validation and sampling techniques like bootstrapping.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	What is **Environment in AI**?	CO1	2	2
b.	What is **Fuzzy Logic**?	CO2	2	2
c.	Define **Reinforcement Learning**.	CO3	3	2
d.	Define **RNN**	CO4	2	2
e.	What is **Bootstrapping**?	CO5	3	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain **Environment types with examples (deterministic, stochastic, etc.)**	CO1	2	5

b.	Describe **Structure of Intelligent Agent in detail**.	CO1	4	5
c.	Explain **State Space formulation with water jug problem**.	C01	3	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Derive **Bayes' Theorem and solve a numerical problem**.	CO2	4	5
b.	Explain **Bayesian Networks with conditional dependency**.	CO2	3	5
c.	Describe **Declarative vs Procedural Knowledge representation**.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain **Neural Network model with diagram**.	CO3	2	5
b.	Discuss **Overfitting and Underfitting with solutions**.	CO3	3	5
c.	Explain **Difference between Supervised, Unsupervised and Reinforcement Learning**.	CO3	3	5
Que. 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain **Difference between CNN and RNN with applications**.	CO4	2	5
b.	Describe **K-NN algorithm and its limitations.**	CO4	4	5
c.	Explain **Regression Decision Trees with example**.	C04	2	5
Que. 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Compare **PCA vs LDA**.	CO5	2	5
b.	Explain **Fuzzy K-Means clustering**.	CO5	3	5
c.	Discuss **Cross Validation vs Bootstrapping techniques**.	C05	2	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create